

PHYS 230 - Electricity and Magnetism

Lecture 25

Instructor: Saeed Rastgoo



**UNIVERSITY
OF ALBERTA**

Outline

5 Current, Resistance, and Electromotive Force

- Energy and Power in Circuits (Continued)

6 Direct-Current Circuits

- Resistors in Series and Parallel
- Kirchhoff's Rules

All figures in these slides are from one of the following sources, unless cited otherwise:

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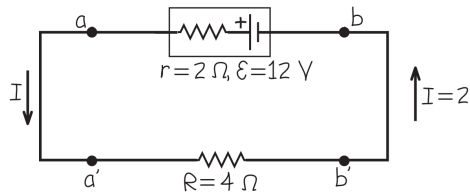
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Exercise I

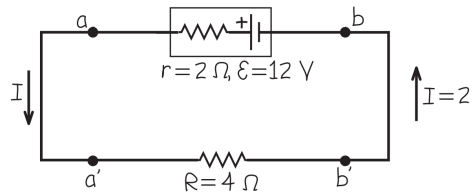
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For the circuit shown, if $V = V_{ba} = V_a - V_b$, find the rates (powers) of the battery's energy conversion, energy dissipation, and net power output, and the rate of energy dissipation in the $4\ \Omega$ resistor



Exercise I

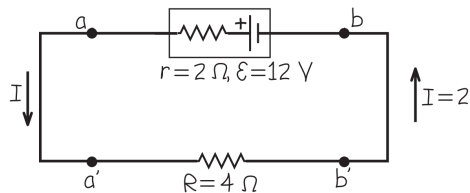
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Exercise 2

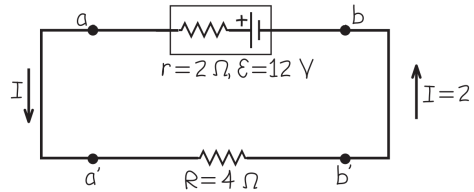
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Suppose we replace the external $4\ \Omega$ resistor in the previous circuit with an $8\ \Omega$ resistor. How does this affect the electrical power dissipated in this resistor?



Exercise 2

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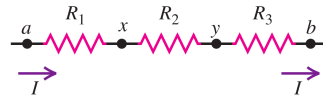
Resistors in Series

Remember from previous lectures: In general, for components in series written in terms of an equivalent component

$$I = I_1 = I_2 = I_3 = \dots \quad (1)$$

$$V = V_1 + V_2 + V_3 + \dots \quad (2)$$

(a) R_1 , R_2 , and R_3 in series



Resistors in Series

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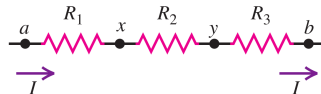
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For the specific case of resistances we have in general

$$V = RI \quad (3)$$

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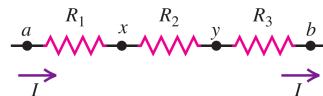
For the specific case of resistances we have in general

$$V = RI \quad (3)$$

Combining equation (2) with equation (3) we get for an equivalent potential in series

$$V = R_1 I_1 + R_2 I_2 + R_3 I_3 + \dots \quad (4)$$

(a) R_1 , R_2 , and R_3 in series



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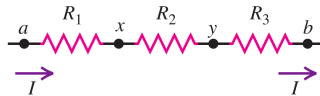
Combining equation (2) with equation (3) we get for an equivalent potential in series

$$V = R_1 I_1 + R_2 I_2 + R_3 I_3 + \dots \quad (4)$$

And now using equation (1) in equation (4) we get

$$V = R_1 I + R_2 I + R_3 I + \dots = \underbrace{(R_1 + R_2 + R_3 + \dots)}_R I$$

(a) R_1 , R_2 , and R_3 in series

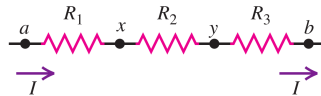


Resistors in Series

So **for an equivalent component in series** with potential V and current I , the **equivalent resistance R** is

$$R = R_1 + R_2 + R_3 + \dots$$

(a) R_1 , R_2 , and R_3 in series



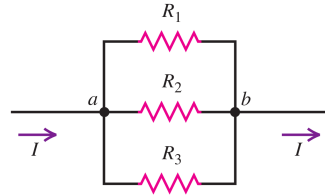
Resistors in Parallel

For components in parallel written in terms of an equivalent component

$$I = I_1 + I_2 + I_3 + \dots \quad (5)$$

$$V = V_1 = V_2 = V_3 = \dots \quad (6)$$

(b) R_1 , R_2 , and R_3 in parallel



Resistors in Parallel

For components in parallel written in terms of an equivalent component

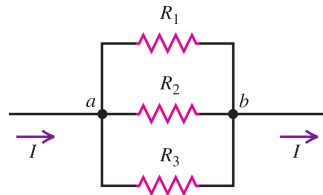
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For the specific case of resistances we have in general

$$V = RI \Rightarrow I = \frac{V}{R} \quad (7)$$

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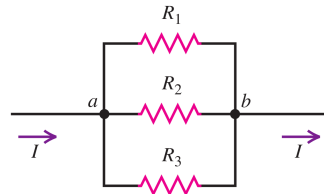
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$$V = RI \Rightarrow I = \frac{V}{R} \quad (7)$$

Combining equation (5) with equation (7) we get for an equivalent potential in series

$$I = \frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} + \dots \quad (8)$$

(b) R_1 , R_2 , and R_3 in parallel



Resistors in Parallel

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$$I = I_1 + I_2 + I_3 + \dots \quad (5)$$

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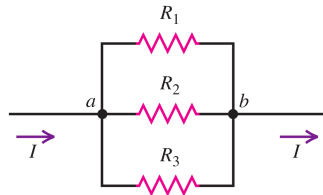
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$$I = \frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} + \dots \quad (8)$$

And now using equation (5) in equation (8) we get

$$I = \frac{V}{R_1} + \frac{V}{R_2} + \frac{V}{R_3} + \dots = V \underbrace{\left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots \right)}_{\frac{1}{R}}$$

(b) R_1 , R_2 , and R_3 in parallel

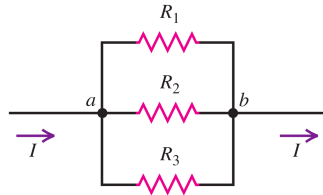


Resistors in Parallel

So for an equivalent component in parallel with potential V and current I , the equivalent resistance R is

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots$$

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Resistors in Parallel

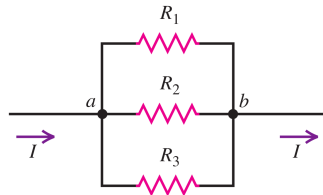
So for an equivalent component in parallel with potential V and current I , the equivalent resistance R is

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots$$

Also since

$$V_1 = V_2 \Rightarrow R_1 I_1 = R_2 I_2 \Rightarrow \frac{I_1}{I_2} = \frac{R_2}{R_1}$$

(b) R_1 , R_2 , and R_3 in parallel

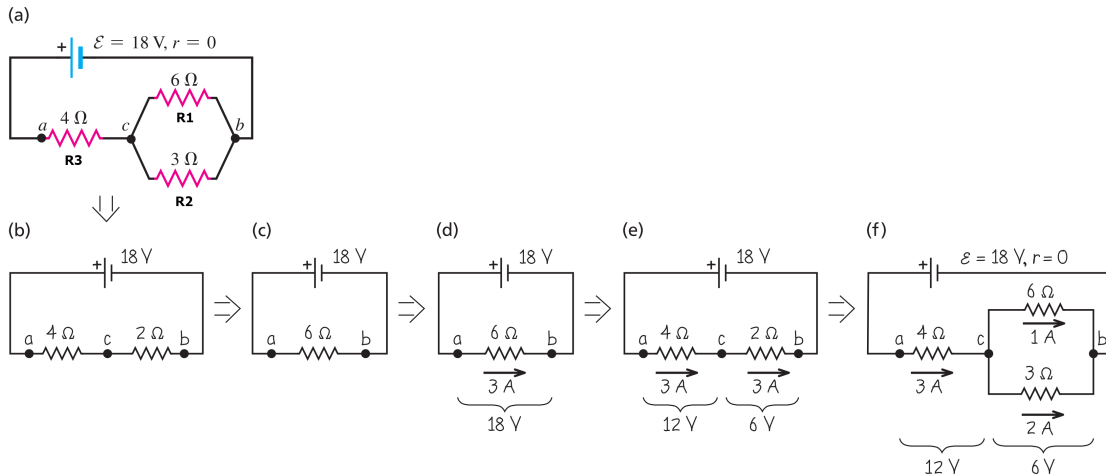


Exercise 3

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Find the equivalent resistance of the network in figure and the current in each resistor. The source of emf has negligible internal resistance.

Figure 26.3 Steps in reducing a combination of resistors to a single equivalent resistor and finding the current in each resistor.



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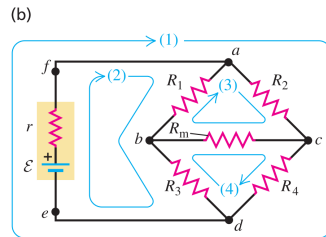
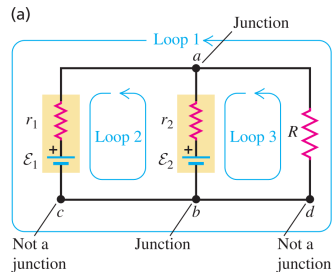
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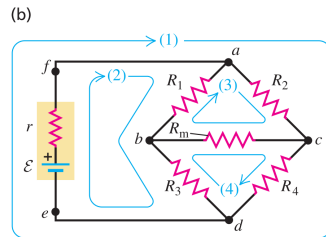
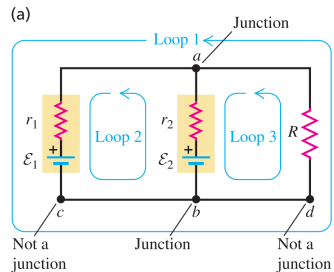
Junctions, Loops, Kirchhoff's Rules

- **Junction:** any point in a circuit where three or more conductors meet.



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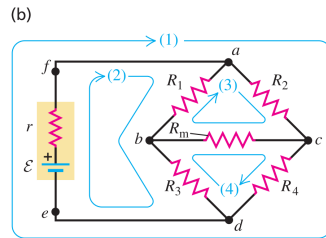
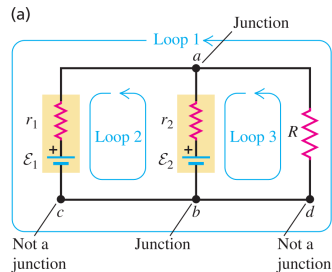
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- **Junction:** any point in a circuit where three or more conductors meet.
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Kirchhoff's junction rule: The sum of the currents into/out of any junction is zero

$$\sum I_n = 0$$

Currents into a junction are positive, I , and currents out of a junction are negative, $-I$



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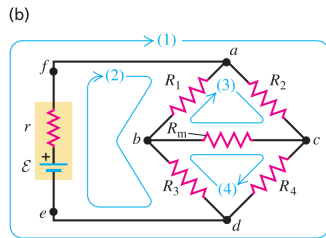
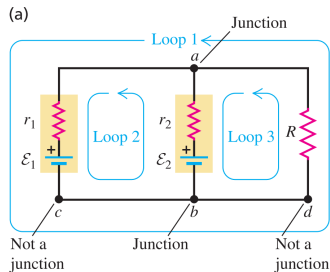
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Kirchhoff's loop rule: The sum of the potential differences around any loop is zero

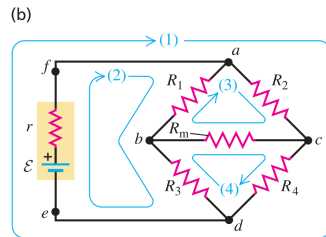
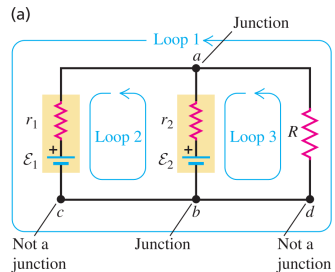
$$\sum V_n = 0$$

Moving in the direction of current, V decreases; moving against the current, V increases



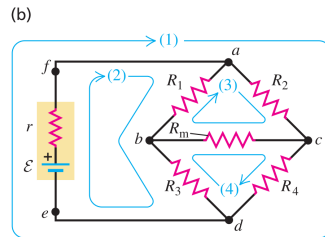
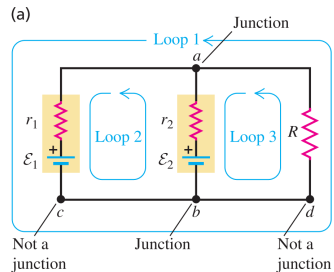
Applying Kirchhoff's Rules

1. Identify all junctions and loops



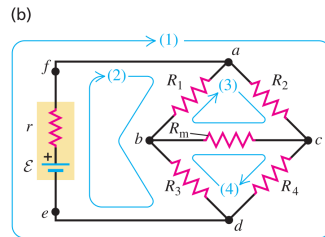
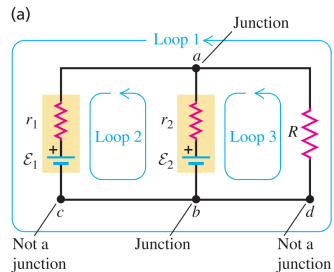
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1. Identify all junctions and loops
2. Assume direction of current in each loop; if you get a negative current at the end, it means you chose the inverse current direction



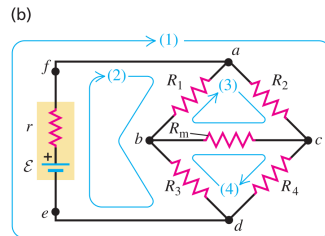
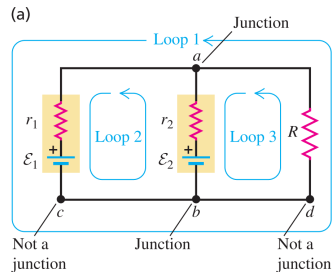
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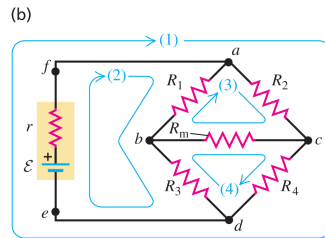
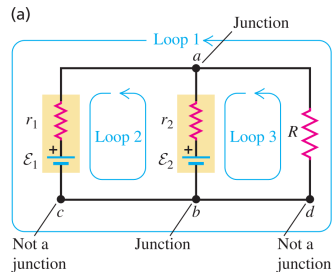
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4. For each loop write the Kirchhoff's loop rule; start from an arbitrary point on the circuit, go around the loop clockwise or counterclockwise:



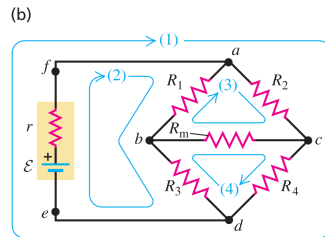
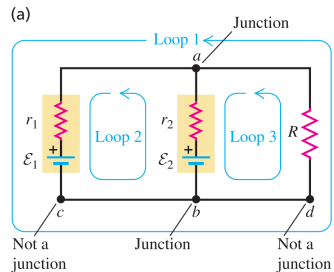
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 - 4.1 Going along the current, subtract the V of the device in the way from previous potentials



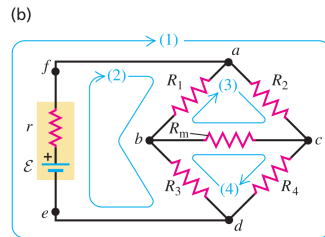
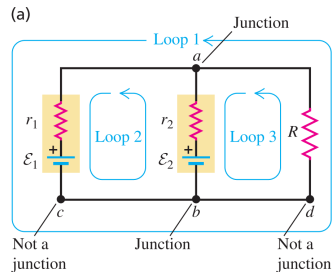
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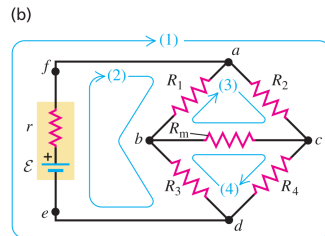
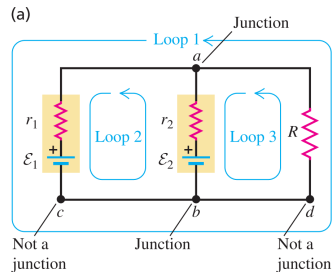
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 - 4.1 Going along the current, subtract the V of the device in the way from previous potentials
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 - 4.3 In batteries, going from $+$ to $-$, subtract the $\mathcal{E}$ of the battery from previous potentials



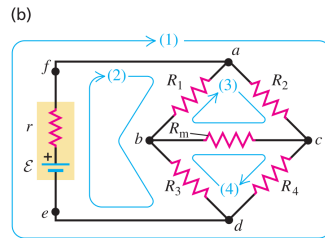
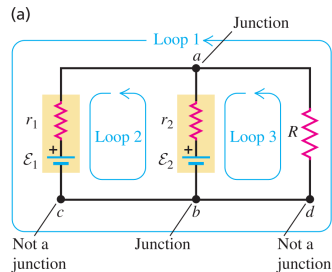
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Applying Kirchhoff's Rules

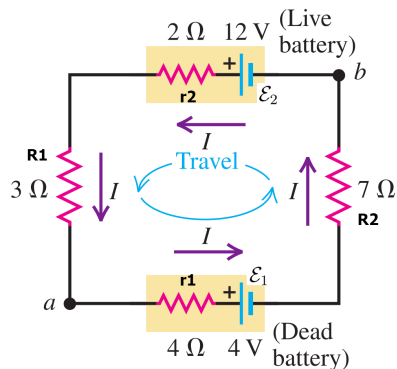
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 - 4.3 In batteries, going from $+$ to $-$, subtract the $\mathcal{E}$ of the battery from previous potentials
 - 4.4 In batteries, going from $-$ to $+$, add the $\mathcal{E}$ of the battery to previous potentials
5. Solve the system of equations from all junction and loop rules together to obtain the unknowns



Exercise 4

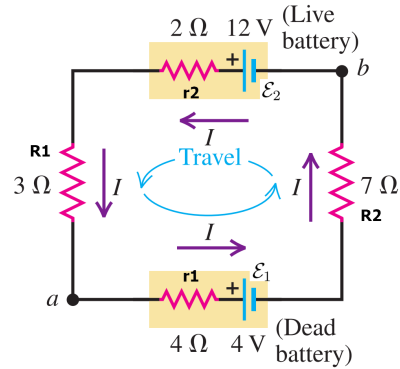
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The circuit shown in figure contains two batteries, each with an emf and an internal resistance, and two resistors. Find (a) the current in the circuit, (b) the potential difference $V_{ba} = V_a - V_b$, and (c) the power output of the emf of each battery.



Exercise 4

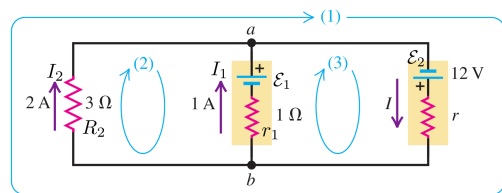
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Exercise 5

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In the circuit shown in figure, a 12 V power supply with unknown internal resistance r is connected to a run-down rechargeable battery with unknown emf $\mathcal{E}_1$ and internal resistance $1\ \Omega$ and to an indicator light bulb of resistance $3\ \Omega$ carrying a current of 2 A . The current through the run-down battery is 1 A in the direction shown. Find r , $\mathcal{E}_1$, and the current I through the power supply.



Exercise 5

<https://epoll.srv.ualberta.ca> – Code: MBD

